

Logarithmic q -Richardson Acceleration and Lambert Phase Locking in Fabius–Rvachev Analysis

Repository-wide audit, exact sinc-tail extrapolation,
Bell-polynomial error calculus, and extraction of the periodic endpoint factor

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Research status. The sinc-product results in Parts I–III are proved from first principles in this report. The Lambert-lattice extrapolation theorem is an algebraic consequence of the all-orders lower-Lambert expansion developed in the repository; its use is therefore explicitly conditional on that expansion with the stated uniformity. The numerical tables were independently recomputed at 100-digit precision. “New” means absent as an exact result from the audited repository snapshot; no claim of worldwide priority is made.

Abstract

The Fabius function, Rvachev's compactly supported C^∞ up-function, the signed Thue–Morse sequence, and dyadic q -calculus form a tightly connected system. The existing PROVEIT corpus already contains extensive exact dyadic arithmetic, Gaussian-binomial interpolation, moment and cumulant formulas involving Bernoulli and Bell polynomials, Fourier products, Lambert- W endpoint asymptotics with a nonconstant periodic factor, and several forms of Richardson extrapolation. After a repository-wide audit of the living documents, frozen historical versions, research drafts, and bundled source papers, this report develops two extrapolation mechanisms that are not stated there.

First, for the generalized lacunary sinc product

$$P_b(t) = \prod_{k \geq 1} \text{sinc}(t/b^k), \quad b > 1,$$

we obtain an exact Bernoulli–zeta expansion of the omitted logarithmic tail. Applying Lagrange interpolation to the logarithms of the truncations at the geometric nodes q^j , $q = b^{-2}$, gives a multiplicative Richardson transform with closed Gaussian- q -binomial weights. Its full error series is exact, has a fixed sign $(-1)^r$ at level r , yields consecutive certified upper and lower bounds, and admits an explicit nonasymptotic remainder estimate. The ℓ^1 norm of the weights is exactly $(-q; q)_r / (q; q)_r$, hence remains bounded as the order tends to infinity; in the dyadic case it stays below 1.970. The method therefore combines high order with unusually mild conditioning. Bell polynomials convert the logarithmic error into relative-error and derivative formulas, while a finite Thue–Morse exponential-sum identity transfers the accelerated product to purely finite signed digital sums.

Second, the exact inverse coordinate

$$\lambda(x) = -\frac{1}{\log 2} W_{-1}(-x \log 2), \quad x = \lambda 2^{-\lambda},$$

is used to build phase-locked nodes $x_j = (\lambda + j)2^{-(\lambda+j)}$. Periodic coefficient functions are identical at all these nodes. Ordinary Lagrange extrapolation in $h_j = (\lambda + j)^{-1}$ then cancels the first r inverse- λ corrections and extracts the periodic endpoint term $\Psi(\lambda)$ directly from values of F . The exact weights, all transformed moments, and their complete-homogeneous/Bell-polynomial representation are derived. Bernoulli polynomials describe the arithmetic-node power sums needed in conditioning and preconditioning. Together these results provide a new bridge between q -binomial interpolation, sinc products, Thue–Morse blocks, Lambert- W phase geometry, Bernoulli coefficients, Bell-polynomial reconstruction, and Richardson extrapolation.

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1 Scope, corpus audit, and nonduplication boundary

1.1 The audited corpus

The target tree contains living expositions, a deliberately quarantined frontier volume, active drafts, frozen historical versions, and source-paper transcriptions. The audit used the current tree and the archive ledger as of 27 August 2026. Counting each distinct `.tex` path, the inventory contains 57 sources: 25 in the living tree and 32 frozen historical sources. The living set consists of four top-level expositions, the canonical frontier volume, one Thue–Morse atlas, twelve Fourier-decay dossiers, and seven source-paper transcriptions.

Table 1: Repository-wide source inventory used for the overlap audit.

Family	TeX files	Principal content relevant to this report
Living primary expositions	4	Fabius/up synthesis, glossary, non-elementarity, Lean walk-through
Canonical frontier volume	1	Dyadic web, repeated integration, endpoint asymptotics, Thue–Morse convergence, saddle analysis
Living formula atlas	1	Thue–Morse identities, Prouhet cancellation, rarefaction, Mahler structure, Bell/Bernoulli formulas
Living Fourier-decay dossiers	12	Infinite sinc products, zeros, dyadic renormalization, envelopes, Lambert- W scales, competing asymptotic formulations
Bundled source-paper TeX	7	Lacunary products, up-function background, arithmetic of F , approximation literature
Frozen archive	32	Earlier notes, parallel expositions, q -calculus articles, supplements, formula atlases, topic dossiers
Total	57	Every path is listed in Appendix A.

The archive matters for originality. Several ideas no longer visible in the living primary paper survive in an older parallel draft or a superseded dossier. In particular, the audit treated the following as prior art rather than rediscoveries:

- Gaussian- q -binomial coefficient extraction at geometric nodes;
- exact dyadic Richardson transforms for sequences modeled by powers of 2^{-p} ;
- Bernoulli-number cumulants and Bell-polynomial moment reconstruction;
- Prouhet cancellation for signed Thue–Morse blocks;
- fixed-numerator Lambert phase locking along dyadic endpoint rays;
- the periodic factor in the corrected small-argument asymptotic;
- exact finite Thue–Morse products and Bromwich/coefficient representations;
- global Fourier-decay envelopes and the zero geometry of the up transform.

1.2 What is new relative to the repository

The exact constructions below were not found in any audited source:

1. Richardson extrapolation of the *logarithm* of truncated lacunary sinc products, followed by exponentiation to a multiplicative approximation;
2. the exact Gaussian- q -binomial formula for those log-extrapolation weights together with the full transformed Bernoulli-zeta tail;
3. the sign theorem and resulting alternating certified brackets for the infinite product;
4. the exact bounded condition number $(-q; q)_r / (q; q)_r$ of the level- r log transform;
5. the Bell-polynomial relative-error and derivative hierarchy of the accelerated product;
6. transfer of this accelerated approximation to finite signed Thue-Morse exponential sums;
7. Lambert phase-locked Richardson nodes $x_j = (\lambda + j)2^{-(\lambda+j)}$ designed specifically to extract the periodic endpoint term Ψ ;
8. exact transformed inverse-power moments on that lattice, including their complete-homogeneous and Bell-polynomial form.

A targeted literature search located general Richardson-extrapolation theory and the standard ingredients separately, but no exact counterpart of the combined sinc-tail construction. The responsible statement is therefore “new relative to the audited repository and apparently not standard,” not a claim of global priority.

1.3 Proof-status convention

Every substantial assertion is tagged implicitly by its placement:

- Results in Sections 3–7 are **proved here**.
- The phase-locked Lambert transform in Section 10 is **proved conditionally on the repository’s all-orders expansion**: the interpolation algebra is proved here, while the all-orders endpoint input is inherited from the formally verified endpoint theory.
- Items in Section 15 are explicitly conjectural or open.

Here “proved” means proved in this mathematical report, not necessarily formalized in Lean. In particular, the endpoint input is represented by `fabiusSaddleLogCoefficient`, `log_fabius_sub_sharpLambertMain_hasAsymptoticExpansion`, and `log_fabius_sub_sharpLambertExpansion_isBigO` in the Lean development, whereas the sinc-tail and phase-extraction transforms remain formalization obligations.

2 Notation and structural background

2.1 Sinc products and the Rvachev transform

We use the entire normalized sinc function

$$\operatorname{sinc} z = \begin{cases} \sin z/z, & z \neq 0, \\ 1, & z = 0. \end{cases}$$

For $b > 1$, define

$$P_b(t) = \prod_{k=1}^{\infty} \operatorname{sinc}\left(\frac{t}{b^k}\right), \quad P_{b,N}(t) = \prod_{k=1}^N \operatorname{sinc}\left(\frac{t}{b^k}\right). \quad (1)$$

The dyadic case is $P = P_2$. In a common Fourier normalization for Rvachev's up-function,

$$\widehat{\text{up}}(\xi) = \prod_{h=0}^{\infty} \text{sinc}\left(\frac{\pi\xi}{2^h}\right) = \text{sinc}(\pi\xi)P_2(\pi\xi). \quad (2)$$

Thus every result for P_2 immediately applies to the nontrivial tail of $\widehat{\text{up}}$.

Set

$$q = b^{-2} \in (0, 1), \quad z_N = q^{N+1}, \quad \rho_N = \frac{|t|}{b^{N+1}} = |t|z_N^{1/2}. \quad (3)$$

The local analysis below requires $\rho_N < \pi$, exactly the condition that every omitted sinc factor lies before its first nonzero zero.

2.2 q -notation

For $r \in \mathbb{N}$,

$$(a; q)_r = \prod_{k=0}^{r-1} (1 - aq^k), \quad (q; q)_r = \prod_{k=1}^r (1 - q^k),$$

and

$$\begin{bmatrix} r \\ j \end{bmatrix}_q = \frac{(q; q)_r}{(q; q)_j (q; q)_{r-j}} \quad (0 \leq j \leq r).$$

We will repeatedly use the finite q -binomial theorem

$$\prod_{k=0}^{r-1} (1 + zq^k) = \sum_{j=0}^r q^{\binom{j}{2}} \begin{bmatrix} r \\ j \end{bmatrix}_q z^j. \quad (4)$$

2.3 Bernoulli and Bell polynomials

The Bernoulli polynomials are defined by

$$\frac{te^{xt}}{e^t - 1} = \sum_{n=0}^{\infty} B_n(x) \frac{t^n}{n!}, \quad B_n = B_n(0). \quad (5)$$

The complete exponential Bell polynomials $\mathcal{B}_n$ satisfy

$$\exp\left(\sum_{m \geq 1} x_m \frac{u^m}{m!}\right) = \sum_{n \geq 0} \mathcal{B}_n(x_1, \dots, x_n) \frac{u^n}{n!}. \quad (6)$$

They are the natural conversion device from logarithmic/cumulant data to multiplicative/moment data.

2.4 The lower-Lambert coordinate

Let $L = \log 2$. For sufficiently small $x > 0$, set

$$\lambda(x) = -\frac{1}{L} W_{-1}(-Lx). \quad (7)$$

Then

$$x = \lambda 2^{-\lambda}, \quad \lambda > \frac{1}{L}. \quad (8)$$

The lower branch is essential: on $\lambda > 1/L$, the map $\lambda \mapsto \lambda 2^{-\lambda}$ is strictly decreasing and hence invertible.

Part I

Exact logarithmic tails and multiplicative q -Richardson acceleration

3 The exact Bernoulli–zeta tail of a lacunary sinc product

3.1 The logarithmic sinc series

Lemma 3.1 (Bernoulli–zeta expansion). *For $|u| < \pi$,*

$$\log \frac{\sin u}{u} = - \sum_{m=1}^{\infty} \gamma_m u^{2m}, \quad \gamma_m = \frac{\zeta(2m)}{m\pi^{2m}} = \frac{2^{2m-1}|B_{2m}|}{m(2m)!}. \quad (9)$$

Every γ_m is strictly positive.

Proof. Euler's product $\sin u/u = \prod_{n \geq 1} (1 - u^2/(\pi^2 n^2))$ is locally uniformly convergent for $|u| < \pi$. Expanding $\log(1 - v) = -\sum_{m \geq 1} v^m/m$ and interchanging the absolutely convergent sums gives

$$\log \frac{\sin u}{u} = - \sum_{m \geq 1} \frac{u^{2m}}{m\pi^{2m}} \sum_{n \geq 1} n^{-2m}.$$

The Bernoulli form follows from Euler's evaluation of $\zeta(2m)$. □

The first coefficients are

$$\log \operatorname{sinc} u = -\frac{u^2}{6} - \frac{u^4}{180} - \frac{u^6}{2835} - \frac{u^8}{37800} - \cdots. \quad (10)$$

3.2 Exact omitted-tail formula

Theorem 3.2 (Exact logarithmic tail). *Let $b > 1$, $N \geq 0$, and $t \in \mathbb{R}$ satisfy $\rho_N < \pi$. If $P_{b,N}(t) \neq 0$, then $P_b(t)$ and $P_{b,N}(t)$ have the same sign and*

$$\log |P_{b,N}(t)| - \log |P_b(t)| = \sum_{m=1}^{\infty} d_m(t) z_N^m, \quad (11)$$

where

$$d_m(t) = \frac{\gamma_m t^{2m}}{1 - q^m} = \frac{\zeta(2m) t^{2m}}{m\pi^{2m}(1 - q^m)}. \quad (12)$$

Equivalently,

$$\log |P_{b,N}(t)| - \log |P_b(t)| = \sum_{m=1}^{\infty} \frac{\zeta(2m)}{m\pi^{2m}(1 - q^m)} \rho_N^{2m}. \quad (13)$$

The series is absolutely convergent and strictly positive for $t \neq 0$.

Proof. For $k \geq N + 1$ we have $|t|/b^k \leq \rho_N < \pi$, so every omitted sinc factor is positive. Moreover, the logarithms of the omitted factors are absolutely summable by [lemma 3.1](#); their product therefore converges to a strictly positive, nonzero limit. Hence truncation and limit have the same sign. Applying the same expansion and using absolute convergence,

$$\begin{aligned} \log |P_b(t)| - \log |P_{b,N}(t)| &= \sum_{k=N+1}^{\infty} \log \operatorname{sinc}(t/b^k) \\ &= - \sum_{m=1}^{\infty} \gamma_m t^{2m} \sum_{k=N+1}^{\infty} b^{-2mk} \\ &= - \sum_{m=1}^{\infty} \gamma_m t^{2m} \frac{q^{m(N+1)}}{1 - q^m}. \end{aligned}$$

Negating yields the result. $\square$

Corollary 3.3 (Analytic Bernoulli correction). *For $M \geq 0$, define*

$$C_{b,N}^{[M]}(t) = P_{b,N}(t) \exp \left(- \sum_{m=1}^M d_m(t) z_N^m \right). \quad (14)$$

Then $C_{b,N}^{[M]}(t)$ has the same sign as $P_b(t)$ and

$$\log \frac{|C_{b,N}^{[M]}(t)|}{|P_b(t)|} = \sum_{m=M+1}^{\infty} d_m(t) z_N^m > 0 \quad (t \neq 0). \quad (15)$$

Thus $|C_{b,N}^{[M]}(t)|$ is a rigorous upper bound for $|P_b(t)|$.

This correction is explicit but uses $\zeta(2m)$ or Bernoulli numbers. The next construction achieves cancellation from product values alone.

4 Geometric Lagrange weights and their q -binomial form

4.1 Interpolation at $1, q, \dots, q^r$

For an integer $r \geq 0$, let $\lambda_{r,j}(q)$ be the Lagrange basis weight for evaluation at 0 using nodes $q^0, \dots, q^r$:

$$\lambda_{r,j}(q) = \prod_{\substack{0 \leq k \leq r \\ k \neq j}} \frac{-q^k}{q^j - q^k}. \quad (16)$$

Theorem 4.1 (Gaussian- q -binomial weights). *For $0 \leq j \leq r$,*

$$\lambda_{r,j}(q) = \frac{(-1)^{r-j} q^{\binom{r-j+1}{2}}}{(q; q)_j (q; q)_{r-j}} = \frac{(-1)^{r-j} q^{\binom{r-j+1}{2}}}{(q; q)_r} \begin{bmatrix} r \\ j \end{bmatrix}_q. \quad (17)$$

Their power moments are

$$S_{r,m}(q) := \sum_{j=0}^r \lambda_{r,j}(q) q^{jm} = \frac{q^{rm} (q^{1-m}; q)_r}{(q; q)_r}. \quad (18)$$

Consequently,

$$S_{r,0} = 1, \quad S_{r,m} = 0 \quad (1 \leq m \leq r), \quad (19)$$

and, for $m \geq r + 1$,

$$S_{r,m}(q) = (-1)^r q^{r(r+1)/2} \frac{(q^{m-r}; q)_r}{(q; q)_r}. \quad (20)$$

In particular, every nonzero residual moment has sign $(-1)^r$.

Proof. Splitting the product in [equation \(16\)](#) into $k < j$ and $k > j$ gives [equation \(17\)](#). For the moments, substitute the Gaussian-binomial form and change variables $k = r - j$. The finite q -binomial theorem [equation \(4\)](#), with a suitable monomial factor extracted, gives [equation \(18\)](#). The zeros for $1 \leq m \leq r$ arise because $(q^{1-m}; q)_r$ contains the factor $1 - q^0$. For $m \geq r + 1$, reverse the negative powers in the product to obtain [equation \(20\)](#). $\square$

4.2 An exact condition number

The alternating weights might appear dangerous, but their total variation is exceptionally well behaved.

Theorem 4.2 (Bounded ℓ^1 norm). *The absolute weight sum is*

$$\kappa_r(q) := \sum_{j=0}^r |\lambda_{r,j}(q)| = \frac{(-q; q)_r}{(q; q)_r} = \prod_{k=1}^r \frac{1 + q^k}{1 - q^k}. \quad (21)$$

Hence

$$\kappa_r(q) \nearrow \kappa_\infty(q) := \frac{(-q; q)_\infty}{(q; q)_\infty} < \infty. \quad (22)$$

For $q = 1/4$,

$$\kappa_\infty(1/4) = 1.9692603536682694319 \dots \quad (23)$$

Proof. Set $k = r - j$ in the absolute-value sum and use [equation \(17\)](#):

$$\kappa_r(q) = \frac{1}{(q; q)_r} \sum_{k=0}^r q^{k(k+1)/2} \begin{bmatrix} r \\ k \end{bmatrix}_q.$$

[Equation \(4\)](#) with $z = q$ identifies the numerator as $\prod_{i=0}^{r-1} (1 + q^{i+1}) = (-q; q)_r$. The infinite product converges because $\sum q^k < \infty$. $\square$

Remark 4.3. This boundedness is not generic for high-order polynomial extrapolation. It is a special consequence of the geometric nodes and is one reason to extrapolate the logarithms: the inputs are of comparable scale, while the transformation itself has uniformly bounded absolute sensitivity.

5 Multiplicative q -Richardson extrapolation

5.1 Definition and exact transformed error

Assume $\rho_N < \pi$ and $P_{b,N}(t) \neq 0$. Define

$$\widehat{P}_{b;N,r}(t) = \operatorname{sgn}(P_{b,N}(t)) \exp \left(\sum_{j=0}^r \lambda_{r,j}(q) \log |P_{b,N+j}(t)| \right). \quad (24)$$

Equivalently,

$$|\widehat{P}_{b;N,r}(t)| = \prod_{j=0}^r |P_{b;N+j}(t)|^{\lambda_{r,j}(q)}. \quad (25)$$

Negative exponents are harmless because the relevant truncations are nonzero.

Theorem 5.1 (Exact multiplicative extrapolation error). *Under the preceding hypotheses,*

$$E_{b;N,r}(t) := \log \frac{|\widehat{P}_{b;N,r}(t)|}{|P_b(t)|} \quad (26)$$

$$= \sum_{m=r+1}^{\infty} d_m(t) z_N^m S_{r,m}(q) \quad (27)$$

$$= \frac{(-1)^r q^{r(r+1)/2}}{(q; q)_r} \sum_{m=r+1}^{\infty} \frac{\zeta(2m)}{m\pi^{2m}} \frac{(q^{m-r}; q)_r}{1 - q^m} \rho_N^{2m}. \quad (28)$$

The series is absolutely convergent. If $t \neq 0$, then

$$\operatorname{sgn} E_{b;N,r}(t) = (-1)^r. \quad (29)$$

Proof. Apply [theorem 3.2](#) to each truncation $N + j$:

$$\log |P_{b;N+j}| = \log |P_b| + \sum_{m \geq 1} d_m z_N^m q^{jm}.$$

The constant term survives because $S_{r,0} = 1$, while the first r powers vanish by [equation \(19\)](#). This proves [equation \(27\)](#). [Equation \(28\)](#) follows from [equation \(20\)](#). Every factor in its sum is positive, proving the sign statement. $\square$

5.2 Certified alternating brackets

Corollary 5.2 (Rigorous upper and lower bounds). *For $t \neq 0$ under the hypotheses of [theorem 5.1](#),*

$$|\widehat{P}_{b;N,2s+1}(t)| < |P_b(t)| < |\widehat{P}_{b;N,2s}(t)| \quad (s \geq 0). \quad (30)$$

Thus any two consecutive extrapolation levels give a certified enclosure of the infinite product. In particular,

$$\left| \log \frac{|\widehat{P}_{b;N,r}(t)|}{|P_b(t)|} \right| \leq \left| \log \frac{|\widehat{P}_{b;N,r}(t)|}{|\widehat{P}_{b;N,r+1}(t)|} \right| \quad (31)$$

whenever the two levels lie on opposite sides as in [equation \(30\)](#).

The bracket is stronger than a mere asymptotic claim: it follows from the exact sign of every term in the transformed tail.

5.3 Leading term and explicit remainder bound

Corollary 5.3 (Sharp leading error). *As $\rho_N \rightarrow 0$ with r fixed,*

$$E_{b;N,r}(t) = C_r(q) \rho_N^{2r+2} + O(\rho_N^{2r+4}), \quad (32)$$

where

$$C_r(q) = (-1)^r q^{r(r+1)/2} \frac{\zeta(2r+2)}{(r+1)\pi^{2r+2}(1-q^{r+1})}. \quad (33)$$

For the dyadic case $q = 1/4$,

$$C_0 = \frac{2}{9}, \quad C_1 = -\frac{1}{675}, \quad C_2 = \frac{1}{178605}, \quad C_3 = -\frac{1}{154224000}. \quad (34)$$

Theorem 5.4 (Nonasymptotic error bound). *Let $a = (\rho_N/\pi)^2 < 1$. Then*

$$|E_{b;N,r}(t)| \leq \frac{q^{r(r+1)/2} \zeta(2)}{(q; q)_r (1-q)(r+1)} \frac{a^{r+1}}{1-a}. \quad (35)$$

Consequently, for every fixed admissible pair (t, N) ,

$$\widehat{P}_{b;N,r}(t) \longrightarrow P_b(t) \quad (r \rightarrow \infty) \quad (36)$$

with a supergeometric factor $q^{r(r+1)/2}$.

Proof. For $m \geq r+1$, [equation \(20\)](#) gives

$$|S_{r,m}(q)| \leq \frac{q^{r(r+1)/2}}{(q; q)_r}.$$

Also $\zeta(2m) \leq \zeta(2)$, $m^{-1} \leq (r+1)^{-1}$, and $(1-q^m)^{-1} \leq (1-q)^{-1}$. Sum the resulting geometric series in a . Since $(q; q)_r$ tends to the positive limit $(q; q)_\infty$, the right-hand side tends to zero faster than $q^{r^2/2}$ times an exponential factor. $\square$

5.4 Richardson table and multiplicative recurrence

The transform can be evaluated without forming the closed weights.

Algorithm 5.5 (Logarithmic q -Richardson table). Set

$$R_{N,0}(t) = \log |P_{b,N}(t)|.$$

For $s \geq 1$, recursively define

$$R_{N,s}(t) = \frac{R_{N+1,s-1}(t) - q^s R_{N,s-1}(t)}{1 - q^s}. \quad (37)$$

Then

$$R_{N,r}(t) = \sum_{j=0}^r \lambda_{r,j}(q) \log |P_{b,N+j}(t)|, \quad \widehat{P}_{b;N,r} = \operatorname{sgn}(P_{b,N}) e^{R_{N,r}}. \quad (38)$$

In multiplicative form,

$$|\widehat{P}_{b;N,s}| = \left(\frac{|\widehat{P}_{b;N+1,s-1}|}{|\widehat{P}_{b;N,s-1}|^{q^s}} \right)^{1/(1-q^s)}. \quad (39)$$

The logarithmic implementation is preferable because it avoids underflow in the rapidly decaying Fourier regime.

5.5 Explicit dyadic levels

For $b = 2$, $q = 1/4$. The first transforms are

$$|\hat{P}_{N,1}| = |P_N|^{-1/3} |P_{N+1}|^{4/3}, \quad (40)$$

$$|\hat{P}_{N,2}| = |P_N|^{1/45} |P_{N+1}|^{-4/9} |P_{N+2}|^{64/45}, \quad (41)$$

$$|\hat{P}_{N,3}| = |P_N|^{-1/2835} |P_{N+1}|^{4/135} |P_{N+2}|^{-64/135} |P_{N+3}|^{4096/2835}. \quad (42)$$

The log errors begin respectively at orders ρ_N^4 , ρ_N^6 , and ρ_N^8 .

5.6 Hybrid Bernoulli–Richardson correction

The exact tail coefficients and the data-driven transform can be combined.

Corollary 5.6 (Hybrid correction). *For $s \geq 0$, define*

$$\hat{P}_{b;N,r}^{[s]}(t) = \hat{P}_{b;N,r}(t) \exp\left(-\sum_{m=r+1}^{r+s} d_m(t) z_N^m S_{r,m}(q)\right). \quad (43)$$

Then

$$\log \frac{|\hat{P}_{b;N,r}^{[s]}(t)|}{|P_b(t)|} = \sum_{m=r+s+1}^{\infty} d_m(t) z_N^m S_{r,m}(q), \quad (44)$$

which still has sign $(-1)^r$. Thus the hybrid retains certified bracketing while moving the first nonzero term from degree $r + 1$ to degree $r + s + 1$.

This is the most direct point at which Bernoulli numbers and Gaussian q -binomial weights cooperate in one exact numerical scheme.

6 Bell-polynomial error calculus and derivatives

6.1 From logarithmic error to relative error

Write

$$e_m^{(r)}(t) = d_m(t) S_{r,m}(q), \quad e_m^{(r)} = 0 \quad (1 \leq m \leq r). \quad (45)$$

Then [equation \(27\)](#) is $E = \sum_{m \geq 1} e_m^{(r)} z_N^m$. Exponentiation and [equation \(6\)](#) give an exact convergent expansion.

Theorem 6.1 (Bell-polynomial relative-error expansion). *Under the hypotheses of [theorem 5.1](#),*

$$\frac{|\hat{P}_{b;N,r}(t)|}{|P_b(t)|} = 1 + \sum_{n=r+1}^{\infty} \frac{1}{n!} \mathcal{B}_n\left(1!e_1^{(r)}, 2!e_2^{(r)}, \dots, n!e_n^{(r)}\right) z_N^n. \quad (46)$$

In particular, because $e_1^{(r)} = \dots = e_r^{(r)} = 0$, the first relative-error coefficient equals the first logarithmic-error coefficient:

$$\frac{|\hat{P}_{b;N,r}|}{|P_b|} - 1 = e_{r+1}^{(r)} z_N^{r+1} + O(z_N^{r+2}). \quad (47)$$

The next coefficients exhibit the familiar cumulant-to-moment pattern. For example,

$$\begin{aligned} [z_N^{r+2}] (e^E - 1) &= e_{r+2}^{(r)} & (r \geq 1), \\ [z_N^{2r+2}] (e^E - 1) &= e_{2r+2}^{(r)} + \frac{1}{2} (e_{r+1}^{(r)})^2. \end{aligned}$$

6.2 Logarithmic derivatives

On an interval avoiding zeros, differentiation of [equation \(11\)](#) is legitimate termwise. Let

$$D^\ell = \frac{d^\ell}{dt^\ell}, \quad (2m)_\ell = 2m(2m-1) \cdots (2m-\ell+1).$$

Theorem 6.2 (Derivative acceleration). *For every fixed $\ell \geq 0$ and admissible t away from zeros,*

$$D^\ell \left(\sum_{j=0}^r \lambda_{r,j} \log |P_{b,N+j}(t)| - \log |P_b(t)| \right) = \sum_{m \geq \max(r+1, \lceil \ell/2 \rceil)} \frac{\gamma_m(2m)_\ell t^{2m-\ell}}{1-q^m} z_N^m S_{r,m}(q). \quad (48)$$

Thus the same q -Richardson table simultaneously cancels the first r geometric tail powers in every logarithmic derivative.

For reconstructing derivatives of the product itself, set $L(t) = \log |P_b(t)|$. Faà di Bruno's formula becomes

$$\frac{P_b^{(n)}(t)}{P_b(t)} = \mathcal{B}_n(L'(t), L''(t), \dots, L^{(n)}(t)) \quad (49)$$

where the sign of P_b is locally constant. Combining [equations \(48\)](#) and [\(49\)](#) gives an accelerated hierarchy for all derivatives of the Fourier transform away from its zeros.

6.3 Application to interzero extrema

Between consecutive zeros of $\widehat{\text{up}}$, an extremum of its modulus satisfies

$$\frac{d}{d\xi} \log |\widehat{\text{up}}(\xi)| = 0. \quad (50)$$

[Equation \(2\)](#) separates the first sinc factor from $P_2(\pi\xi)$. [Theorem 6.2](#) gives a certified high-order approximation to the tail contribution in [equation \(50\)](#). This suggests a numerically stable alternative to multiplying many tiny sinc factors and then differentiating the product.

7 Transfer to finite signed Thue–Morse sums

Let

$$\varepsilon_n = (-1)^{s_2(n)}, \quad (51)$$

where $s_2(n)$ is the binary digit sum. The standard finite product identity

$$\sum_{n=0}^{2^N-1} \varepsilon_n z^n = \prod_{j=0}^{N-1} (1 - z^{2^j}) \quad (52)$$

becomes a precise sinc identity after a phase-scaled substitution.

Proposition 7.1 (Finite Thue–Morse representation of P_N). *Define*

$$T_N(t) = \sum_{n=0}^{2^N-1} \varepsilon_n \exp\left(-\frac{itn}{2^{N-1}}\right). \quad (53)$$

Then

$$T_N(t) = i^N 2^{-N(N-1)/2} e^{-it(1-2^{-N})} t^N P_N(t), \quad (54)$$

with the identity at $t = 0$ understood by continuity. Hence, for $t \neq 0$,

$$P_N(t) = 2^{N(N-1)/2} i^{-N} e^{it(1-2^{-N})} t^{-N} T_N(t). \quad (55)$$

Proof. Insert $z = e^{-it/2^{N-1}}$ in [equation \(52\)](#). Reindex the factors to obtain

$$T_N(t) = \prod_{k=0}^{N-1} (1 - e^{-it/2^k}).$$

Using

$$1 - e^{-ia} = 2ie^{-ia/2} \sin(a/2),$$

sum the phases. Finally, factor

$$\prod_{j=1}^N \sin(t/2^j) = t^N 2^{-N(N+1)/2} P_N(t),$$

which gives the result. □

Corollary 7.2 (Accelerated infinite product from finite digital sums). *For $t \neq 0$ and admissible N ,*

$$|\widehat{P}_{N,r}(t)| = \prod_{j=0}^r \left(2^{(N+j)(N+j-1)/2} |t|^{-(N+j)} |T_{N+j}(t)| \right)^{\lambda_{r,j}(1/4)}. \quad (56)$$

The exact error, bracketing, and bounds of [theorems 5.1 and 5.4](#) and [corollary 5.2](#) therefore become statements purely about a finite collection of signed Thue–Morse exponential sums.

This transfer is conceptually useful even when direct product evaluation is faster. It makes the approximation a finite digital identity and exposes a new route for symbolic or formal treatment through the already developed Thue–Morse polynomial machinery.

8 Numerical verification and observed orders

All values in this section were recomputed with 100 decimal digits. The reference value used 300 sinc factors; at the displayed precision its omitted tail is negligible relative to every table entry. We report absolute logarithmic error

$$\varepsilon_{N,r}(t) = \left| \log |\widehat{P}_{N,r}(t)| - \log |P(t)| \right|,$$

where $r = 0$ denotes the raw truncation P_N .

The successive- N error ratios approach

$$4^{-1}, \quad 4^{-2}, \quad 4^{-3}, \quad 4^{-4}$$

Table 2: Dyadic product at $t = 3$: absolute logarithmic errors.

N	raw $r = 0$	$r = 1$	$r = 2$	$r = 3$
2	$3.1368194 \cdot 10^{-2}$	$2.9611726 \cdot 10^{-5}$	$1.5785787 \cdot 10^{-8}$	$2.5743474 \cdot 10^{-12}$
3	$7.8198398 \cdot 10^{-3}$	$1.8359337 \cdot 10^{-6}$	$2.4411880 \cdot 10^{-10}$	$9.9424663 \cdot 10^{-15}$
4	$1.9535830 \cdot 10^{-3}$	$1.1451700 \cdot 10^{-7}$	$3.8045692 \cdot 10^{-12}$	$3.8728157 \cdot 10^{-17}$
5	$4.8830986 \cdot 10^{-4}$	$7.1537455 \cdot 10^{-9}$	$5.9408270 \cdot 10^{-14}$	$1.5117515 \cdot 10^{-19}$
6	$1.2207210 \cdot 10^{-4}$	$4.4705340 \cdot 10^{-10}$	$9.2810541 \cdot 10^{-16}$	$5.9042379 \cdot 10^{-22}$

Table 3: Errors for two additional arguments, illustrating that the order is independent of t inside the convergence disk.

t	N	raw $r = 0$	$r = 1$	$r = 2$	$r = 3$
1	4	$2.1701954 \cdot 10^{-4}$	$1.4129551 \cdot 10^{-9}$	$5.2149215 \cdot 10^{-15}$	$5.8978483 \cdot 10^{-21}$
1	6	$1.3563390 \cdot 10^{-5}$	$5.5189741 \cdot 10^{-12}$	$1.2730608 \cdot 10^{-18}$	$8.9985206 \cdot 10^{-26}$
5	4	$5.4288846 \cdot 10^{-3}$	$8.8466438 \cdot 10^{-7}$	$8.1669316 \cdot 10^{-11}$	$2.3096377 \cdot 10^{-15}$
5	6	$3.3909800 \cdot 10^{-4}$	$3.4497407 \cdot 10^{-9}$	$1.9894410 \cdot 10^{-14}$	$3.5155982 \cdot 10^{-20}$

for levels $r = 0, 1, 2, 3$, exactly as [equation \(32\)](#) predicts. The signs also alternate: raw and level two are upper approximations in modulus; levels one and three are lower approximations.

For $t = 3$ the product itself is

$$P(3) = 0.585719176535068482912529262451 \dots \quad (57)$$

Even at $N = 2$, the third transformed level improves the raw log error by approximately ten decimal orders while using only truncations through $N + 3 = 5$.

Part II

Lambert phase-locked extraction of the endpoint oscillation

9 The all-orders endpoint template

The repository's corrected endpoint analysis has the structural form

$$\log F(x) = \mathcal{W}_{\text{MSE}}(x) + \Psi(\lambda(x)) + \sum_{m=1}^M \frac{A_m(\lambda(x))}{\lambda(x)^m} + O(\lambda(x)^{-M-1}), \quad (58)$$

where Ψ and the coefficient functions A_m are 1-periodic. The nonoscillatory main saddle exponent may be written in the normalization used in the repository as

$$\mathcal{W}_{\text{MSE}}(x) = -\frac{L}{2}\lambda^2 + \left(1 + \frac{L}{2}\right)\lambda - \frac{1}{2}\log \lambda + C_F, \quad L = \log 2, \quad (59)$$

with the convention-dependent constant C_F . The critical fact for the present construction is not the explicit constant but the periodicity of every A_m in the same phase variable.

Define the reduced endpoint observable

$$Y(x) = \log F(x) - \mathcal{W}_{\text{MSE}}(x). \quad (60)$$

Then [equation \(58\)](#) becomes

$$Y(x) = \Psi(\lambda) + \sum_{m \geq 1} A_m(\lambda) \lambda^{-m} \quad (61)$$

in the asymptotic sense.

10 Phase-locked Lambert nodes

10.1 Exact phase locking

Fix $\lambda > 1/L$ and define

$$\lambda_j = \lambda + j, \quad x_j = x_j(\lambda) = (\lambda + j)2^{-(\lambda+j)}, \quad j = 0, 1, \dots \quad (62)$$

Lemma 10.1 (Exact Lambert inversion on the lattice). *For every $j \geq 0$,*

$$\lambda(x_j) = \lambda + j. \quad (63)$$

Consequently, for every 1-periodic function G ,

$$G(\lambda(x_j)) = G(\lambda). \quad (64)$$

Proof. [Equation \(8\)](#) gives the inverse relation exactly on the decreasing branch. Periodicity gives the second assertion. $\square$

Thus the nodes are not merely close in phase; they have exactly identical phase. Their physical spacing is nongeometric:

$$\frac{x_{j+1}}{x_j} = \frac{\lambda + j + 1}{2(\lambda + j)}, \quad \log_2 \frac{x_j}{x_0} = \log_2 \frac{\lambda + j}{\lambda} - j. \quad (65)$$

This slight deviation from a dyadic ray is precisely what preserves the Lambert phase.

10.2 Lagrange weights in reciprocal scale

Set

$$h_j = \frac{1}{\lambda + j}. \quad (66)$$

We interpolate $Y(x_j)$ as a formal/asymptotic power series in h_j and evaluate at $h = 0$.

Theorem 10.2 (Phase-locked Richardson weights). *The Lagrange weights for evaluation at $h = 0$ from $h_0, \dots, h_r$ are*

$$\omega_{r,j}(\lambda) = \prod_{\substack{0 \leq k \leq r \\ k \neq j}} \frac{-h_k}{h_j - h_k} = \frac{(-1)^{r-j}}{r!} \binom{r}{j} (\lambda + j)^r. \quad (67)$$

They satisfy

$$\sum_{j=0}^r \omega_{r,j} = 1, \quad \sum_{j=0}^r \omega_{r,j} (\lambda + j)^{-m} = 0 \quad (1 \leq m \leq r). \quad (68)$$

Proof. Since

$$h_j - h_k = \frac{k - j}{(\lambda + j)(\lambda + k)},$$

each factor in the Lagrange product equals $(\lambda + j)/(j - k)$. Therefore

$$\omega_{r,j} = \frac{(\lambda + j)^r}{\prod_{k \neq j} (j - k)} = \frac{(-1)^{r-j}}{j!(r-j)!} (\lambda + j)^r.$$

The moment identities follow either from polynomial interpolation or from the fact that an r th finite difference annihilates every polynomial of degree below r . $\square$

The first three levels are

$$(\omega_{1,0}, \omega_{1,1}) = (-\lambda, \lambda + 1), \quad (69)$$

$$(\omega_{2,0}, \omega_{2,1}, \omega_{2,2}) = \left(\frac{\lambda^2}{2}, -(\lambda + 1)^2, \frac{(\lambda + 2)^2}{2} \right), \quad (70)$$

$$(\omega_{3,0}, \omega_{3,1}, \omega_{3,2}, \omega_{3,3}) = \left(-\frac{\lambda^3}{6}, \frac{(\lambda + 1)^3}{2}, -\frac{(\lambda + 2)^3}{2}, \frac{(\lambda + 3)^3}{6} \right). \quad (71)$$

10.3 All residual moments

Let $h_n(a_0, \dots, a_r)$ denote the complete homogeneous symmetric polynomial, characterized by

$$\prod_{j=0}^r (1 - a_j z)^{-1} = \sum_{n=0}^{\infty} h_n(a_0, \dots, a_r) z^n. \quad (72)$$

Theorem 10.3 (Exact inverse-power moments). *For $s \geq 1$,*

$$\sum_{j=0}^r \omega_{r,j} (\lambda + j)^{-(r+s)} = \frac{(-1)^r}{(\lambda)_{r+1}} h_{s-1} \left(\frac{1}{\lambda}, \frac{1}{\lambda + 1}, \dots, \frac{1}{\lambda + r} \right), \quad (73)$$

where $(\lambda)_{r+1} = \lambda(\lambda + 1) \cdots (\lambda + r)$ is the rising factorial. In particular,

$$\sum_{j=0}^r \omega_{r,j} (\lambda + j)^{-(r+1)} = \frac{(-1)^r}{(\lambda)_{r+1}}. \quad (74)$$

Proof. Substituting [equation \(67\)](#) reduces the left-hand side to

$$\frac{1}{r!} \sum_{j=0}^r (-1)^{r-j} \binom{r}{j} (\lambda + j)^{-s}.$$

For $s = 1$, partial fractions of $1/(\lambda)_{r+1}$ give [equation \(74\)](#). Repeated differentiation with respect to λ , or multiplication of the generating function of reciprocal powers, yields [equation \(73\)](#). Equivalently, compare coefficients in

$$\frac{1}{(\lambda - z)_{r+1}} = \frac{1}{(\lambda)_{r+1}} \prod_{j=0}^r \left(1 - \frac{z}{\lambda + j} \right)^{-1}.$$

$\square$

Define generalized harmonic sums

$$H_k^{(r)}(\lambda) = \sum_{j=0}^r (\lambda + j)^{-k}. \quad (75)$$

Then the complete homogeneous polynomial in [equation \(73\)](#) has the Bell representation

$$h_n\left(\frac{1}{\lambda}, \dots, \frac{1}{\lambda + r}\right) = \frac{1}{n!} \mathcal{B}_n\left(0!H_1^{(r)}, 1!H_2^{(r)}, \dots, (n-1)!H_n^{(r)}\right). \quad (76)$$

This is the second exact Bell-polynomial bridge in the report: the first exponentiated a log error; the second resolves all residual moments of the Lambert extrapolator.

10.4 Extraction of the periodic endpoint factor

Define

$$\widehat{\Psi}_r(\lambda) = \sum_{j=0}^r \omega_{r,j}(\lambda) Y(x_j(\lambda)). \quad (77)$$

Theorem 10.4 (Phase-locked extraction of Ψ). *Assume the all-orders expansion [equation \(58\)](#) holds uniformly along a fixed compact phase set. Then, as $\lambda \rightarrow \infty$ with its fractional part fixed,*

$$\widehat{\Psi}_r(\lambda) - \Psi(\lambda) \sim \frac{(-1)^r}{(\lambda)_{r+1}} \sum_{s=0}^{\infty} A_{r+1+s}(\lambda) h_s\left(\frac{1}{\lambda}, \dots, \frac{1}{\lambda + r}\right) \quad (78)$$

$$= \frac{(-1)^r A_{r+1}(\lambda)}{(\lambda)_{r+1}} + O(\lambda^{-r-2}). \quad (79)$$

In particular, the first r inverse- λ corrections are cancelled exactly.

Proof. At the phase-locked nodes, periodicity gives

$$Y(x_j) = \Psi(\lambda) + \sum_{m \geq 1} A_m(\lambda) (\lambda + j)^{-m}.$$

The constant survives because the weights sum to one. Terms $1 \leq m \leq r$ vanish by [equation \(68\)](#). Write $m = r + 1 + s$ and apply [theorem 10.3](#). A small uniformity point is essential: the absolute weight sum is $O_r(\lambda^r)$. To obtain the expansion through residual index $s = S$, apply the input all-orders theorem through order $2r + S + 1$. Its transformed remainder is then $O_r(\lambda^{-r-S-2})$, while the finitely many undisplayed transformed coefficients start at the same order. This proves the displayed Poincaré expansion despite the growing weights. $\square$

Exponentiating produces a multiplicative estimator of the periodic amplitude:

$$\widehat{\mathcal{A}}_r(\lambda) = \exp(\widehat{\Psi}_r(\lambda)) = \prod_{j=0}^r \left(\frac{F(x_j)}{\exp(\mathcal{W}_{\text{MSE}}(x_j))} \right)^{\omega_{r,j}(\lambda)}. \quad (80)$$

Its relative-error coefficients are again complete Bell polynomials applied to the logarithmic coefficients in [equation \(78\)](#).

10.5 Coefficient extraction beyond Ψ

The same nodes can estimate $A_1, A_2, \dots$ by differentiating the interpolating polynomial at $h = 0$. If $\ell_j(h)$ is the Lagrange basis for h_j , then

$$\widehat{A}_s^{(r)}(\lambda) = \frac{1}{s!} \sum_{j=0}^r \ell_j^{(s)}(0) Y(x_j), \quad 0 \leq s \leq r, \quad (81)$$

with $\widehat{A}_0^{(r)} = \widehat{\Psi}_r$. For $s = 1$,

$$\ell_j'(0) = -\omega_{r,j} \sum_{\substack{0 \leq k \leq r \\ k \neq j}} (\lambda + k). \quad (82)$$

Higher derivative weights are elementary symmetric polynomials in the numbers $-(\lambda + k)$ and can be generated by Bell polynomials. Thus a single phase-locked sample block can recover a triangular set of periodic coefficient functions.

11 Bernoulli-polynomial geometry and conditioning of the Lambert lattice

Unlike the geometric q -weights, the phase-locked weights grow polynomially in λ :

$$\kappa_r^L(\lambda) := \sum_{j=0}^r |\omega_{r,j}(\lambda)| = \frac{1}{r!} \sum_{j=0}^r \binom{r}{j} (\lambda + j)^r. \quad (83)$$

Equivalently, if $J \sim \text{Binomial}(r, 1/2)$,

$$\kappa_r^L(\lambda) = \frac{2^r}{r!} \mathbb{E}(\lambda + J)^r \sim_{\lambda \rightarrow \infty, r \text{ fixed}} \frac{2^r}{r!} \lambda^r. \quad (84)$$

Thus high precision is essential. This is not a defect of the derivation: extrapolating from $h_j \asymp 1/\lambda$ to the remote point $h = 0$ is intrinsically ill-conditioned.

The binomial moments can be expanded through Stirling numbers:

$$\sum_{j=0}^r \binom{r}{j} j^k = \sum_{\ell=0}^k \left\{ \begin{matrix} k \\ \ell \end{matrix} \right\} r^\ell 2^{r-\ell}. \quad (85)$$

Here $r^\ell = r(r-1)\cdots(r-\ell+1)$ is a falling factorial. For unweighted arithmetic-node diagnostics, Bernoulli polynomials give the exact power sums

$$\sum_{j=0}^r (\lambda + j)^p = \frac{B_{p+1}(\lambda + r + 1) - B_{p+1}(\lambda)}{p + 1}. \quad (86)$$

Equation (86) is useful for scaling, centering, and constructing orthogonalized polynomial bases on the Lambert node block. It also makes explicit why Bernoulli polynomials are the natural arithmetic-lattice companion to the Bernoulli numbers already present in the sinc-tail coefficients.

11.1 A stable evaluation strategy

For practical extraction of Ψ :

1. compute λ and x_j at precision exceeding the target by roughly $\log_{10} \kappa_r^L(\lambda)$ guard digits;
2. evaluate $Y(x_j)$ with the same phase convention and main term;
3. use barycentric interpolation in h_j rather than explicitly summing large alternating monomials;
4. repeat at successive λ with the same fractional part and verify the predicted $\lambda^{-(r+1)}$ decay;
5. compare r and $r + 1$ transforms as an a posteriori diagnostic, while remembering that no sign bracket is presently available because the signs of $A_m(\lambda)$ are not known globally.

Part III

Unified interpretation, formalization path, and research program

12 Two extrapolation geometries

The two constructions are instances of one principle: choose a coordinate in which the defect has a power expansion, then interpolate to the zero-defect point.

Table 4: Comparison of the two extrapolation geometries.

Feature	Sinc-tail transform	Lambert endpoint transform
Defect coordinate	$z_N = q^{N+1}$	$h = 1/\lambda$
Node family	$z_N q^j$	$(\lambda + j)^{-1}$
Invariant structure	lacunary tail coefficients	periodic phase coefficients
Interpolation algebra	Gaussian q -binomial	ordinary binomial/finite difference
Leading cancellation	powers $z, \dots, z^r$	powers $h, \dots, h^r$
Coefficient source	Bernoulli numbers / $\zeta(2m)$	periodic saddle jets A_m
Reconstruction	exponential Bell polynomials	complete homogeneous and Bell polynomials
Conditioning	uniformly bounded in r	grows like $2^r \lambda^r / r!$
Certification	exact alternating brackets	asymptotic order, no global sign yet

The sinc scheme is unusually favorable because the geometric-node Lagrange weights have bounded total variation. The Lambert scheme pays for exact phase locking with growing weights, but it solves a different problem: extracting a nonconstant periodic amplitude that ordinary dyadic extrapolation would mix across phases.

13 Consequences for Fabius and Rvachev computations

13.1 Certified Fourier evaluation

Using [equation \(2\)](#), compute the first factor exactly and bracket $P_2(\pi\xi)$ by consecutive transformed levels. This produces a certified enclosure for $|\widehat{\text{up}}(\xi)|$ away from zeros. Because the work is done in logarithms, the method remains usable deep in the log-normal decay regime where direct products underflow.

13.2 Peak localization and envelope testing

Use the derivative transform in [equation \(48\)](#) inside [equation \(50\)](#). The resulting roots can test fine envelope conjectures without contaminating the comparison by truncation error. In particular, a claimed subleading Fourier envelope should be compared against a product value whose numerical error is below the proposed correction term; the bracket theorem makes that requirement checkable.

13.3 Independent extraction of the endpoint oscillation

The phase-locked transform [equation \(77\)](#) gives a second way to recover Ψ : not from its Fourier/Gamma-zeta representation, but from F itself. Agreement between the two routes would be a stringent validation of constants, branches, and phase conventions in the endpoint asymptotic.

13.4 A bridge from exact dyadic values to asymptotics

The repository already relates exact dyadic values to a phase-locked Lambert coordinate on fixed-numerator rays. The new construction suggests a complementary experiment:

1. use exact or high-precision values of F at nearby phase-locked non-dyadic nodes x_j ;
2. extract Ψ and several A_m ;
3. compare them with the periodic coefficient functions inferred from exact dyadic sequences whose phases become dense;
4. use the discrepancy to isolate the dyadic-arithmetic contribution from the universal saddle contribution.

This does not by itself prove the exact-to-asymptotic bridge, but it supplies a sharply conditioned diagnostic coordinate for that problem.

13.5 Base- b generalized up-functions

Theorems [3.2–5.4](#) hold for every real $b > 1$. If a compactly supported refinement function has Fourier transform $\prod_{k \geq 1} \text{sinc}(t/b^k)$ or a finite product times this tail, the same acceleration and bracketing apply with $q = b^{-2}$. This broadens the method beyond the dyadic Rvachev function and may be useful for nonbinary infinite convolutions.

14 Lean formalization blueprint

The sinc-side results are well suited to formalization because their analytic and algebraic components separate cleanly.

14.1 Suggested definitions

1. `lacunarySincProd b N t`: finite product over $k = 1, \dots, N$.
2. `lacunarySincProdInf b t`: infinite product, with convergence proved from $\sum t^2 b^{-2k}$.
3. `qRichardsonWeight q r j`: the closed form [equation \(17\)](#).
4. `qRichardsonLog q r f N`: weighted sum of $f(N + j)$.
5. `lambertPhaseNode lambda j`: $(\lambda + j)2^{-(\lambda+j)}$.
6. `lambertRichardsonWeight lambda r j`: [equation \(67\)](#).

14.2 Algebraic theorem chain

The lowest-risk formalization order is:

1. prove the product form of $\lambda_{r,j}$ and its Gaussian-binomial simplification;
2. prove the moment identity [equation \(18\)](#) from the finite q -binomial theorem;
3. prove the absolute weight sum [equation \(21\)](#);
4. prove the Lambert weight formula and cancellation moments;
5. prove the reciprocal moment identity [equation \(73\)](#) by rational-function coefficients;
6. formalize the Bell-polynomial conversions as finite coefficient identities.

These steps require no delicate asymptotics.

14.3 Analytic theorem chain

The analytic side can then proceed as follows:

1. establish the log-sinc power series on $(-\pi, \pi)$;
2. justify the double-series interchange for the omitted tail;
3. derive the exact transformed error by finite linearity;
4. prove positivity of every factor in [equation \(28\)](#);
5. deduce bracketing and the explicit majorant;
6. add termwise differentiation on compact zero-free intervals.

The endpoint input is already formalized with its exact phase and every finite remainder: the periodic coefficient functions are `fabiusSaddleLogCoefficient`, and the expansion is proved in `FabiusFullAsymptoticExpansion.lean`. What remains in the frontier namespace is the phase-locked extraction transform itself and the verification that its weighted use of those uniform remainders has precisely the strength asserted in [theorem 10.4](#). The arbitrary-node interpolation foundation is already available in `GeometricLagrange.lean`; the closed reciprocal-grid weights and their all-residual moments are the next algebraic obligations.

15 Open problems and next frontier steps

15.1 Optimal order under finite precision

For exact input logs, [equation \(35\)](#) improves supergeometrically with r , while [equation \(21\)](#) remains bounded. In floating-point arithmetic, however, the truncations P_{N+j} are themselves computed recursively and share correlated rounding errors. A sharp forward-error theorem should combine:

$$\text{analytic error} \asymp q^{r(r+1)/2} a^{r+1} \quad \text{with} \quad \text{roundoff} \lesssim \kappa_r(q)u.$$

Because $\kappa_r(q)$ converges, the optimal r may be much larger than in ordinary Richardson extrapolation.

15.2 Monotonicity within each bracket parity

The sign theorem places every even level above the limit in modulus and every odd level below it, but it does not order two levels of the same parity. In fact, such an ordering is false over the full parameter range $b > 1$. To see this directly, take

$$q = \frac{4}{5}, \quad b = q^{-1/2} = \frac{\sqrt{5}}{2}, \quad N = 0, \quad \rho_0 = \pi \sqrt{\frac{99}{100}}, \quad t = b\rho_0 = \pi \sqrt{\frac{99}{80}}.$$

This is admissible because $\rho_0 < \pi$, and $P_{b,0} = 1$ is the nonzero empty product. Substitution in the absolutely convergent series [\(28\)](#), with $a = (\rho_0/\pi)^2 = 99/100$, gives

r	0	1	2	3
$E_{b,0,r}(t)$	13.5220607386	-12.9320597704	16.5469385988	-14.7912158217.

Thus $E_{b,0,2} > E_{b,0,0} > 0$ and $E_{b,0,3} < E_{b,0,1} < 0$: both proposed same-parity monotonicities fail. These numbers are reproducible by summing the terms through $m = 5000$ in [\(28\)](#). The same majorization used in [theorem 5.4](#) bounds the discarded tail, for each $0 \leq r \leq 3$, by

$$\frac{q^{r(r+1)/2} \zeta(2)}{(q; q)_r (1-q)(5001)} \frac{a^{5001}}{1-a} < 2 \times 10^{-22}.$$

The correctly parameterized open question is therefore: for which pairs $(q, a) \in (0, 1)^2$, and for which levels s , do

$$E_{2s}(q, a) > E_{2s+2}(q, a) > 0, \quad E_{2s+1}(q, a) < E_{2s+3}(q, a) < 0,$$

where $E_r(q, a)$ denotes the right-hand side of [\(28\)](#) after replacing $(\rho_N/\pi)^2$ by a ? No nonempty parameter regime for these additional inequalities is asserted here. Whenever they do hold, consecutive same-parity levels sharpen the certified brackets of [corollary 5.2](#).

15.3 Global continuation across zeros

The logarithmic construction is naturally local to a zero-free component. A global formulation could factor all explicitly known zeros from the truncations and extrapolate the analytic nonvanishing remainder. Such a theorem would give uniform certified evaluation across large frequency intervals, including neighborhoods of high-multiplicity dyadic zeros.

15.4 Direct formulas for the phase coefficients

The transformed moments [equation \(73\)](#) show exactly how $A_{r+1}, A_{r+2}, \dots$ enter the extraction error. This invites an inverse problem: compute $\widehat{\Psi}_r$ and $\widehat{A}_s^{(r)}$ at many phases, then identify their Fourier coefficients in terms of the Gamma–zeta data already present in the endpoint theory.

15.5 A sign theory for Lambert corrections

A sign pattern for $A_m(\lambda)$ would turn the phase-locked transform into a certified bracket analogous to [corollary 5.2](#). Even a sign theorem for A_1 or A_2 on a substantial phase interval would be useful.

15.6 Thue–Morse arithmetic of transformed products

For $q = 1/4$, every exponent in [equations \(40\) to \(42\)](#) is rational. Raising to a common denominator converts the approximant to a rational monomial identity in finite Thue–Morse sums. It is natural to ask whether the numerators and denominators exhibit systematic 2-adic cancellation analogous to the exact dyadic arithmetic of F .

16 Conclusion

The repository already contains the ingredients of a remarkable analytic-combinatorial machine: Thue–Morse cancellation, Gaussian interpolation, exact dyadic arithmetic, infinite sinc products, Bernoulli cumulants, Bell reconstruction, and Lambert- W endpoint geometry. The new results here arise by changing *what* is extrapolated and *where* the nodes are placed.

For the Fourier product, extrapolating logarithms exposes an exact positive power series in the geometric tail coordinate. Gaussian- q -binomial weights annihilate its first powers, and the remaining q -moments all have one sign. This yields high-order multiplicative approximants, exact alternating brackets, explicit bounds, bounded conditioning, Bell-polynomial relative errors, derivative acceleration, and a finite Thue–Morse realization.

For the endpoint asymptotic, moving by integer steps in the exact lower-Lambert coordinate freezes every periodic coefficient. Lagrange interpolation in reciprocal λ then isolates the periodic factor and its correction hierarchy. The residual moments are not opaque: they are complete homogeneous symmetric polynomials, hence Bell polynomials in generalized harmonic sums, while Bernoulli polynomials govern the arithmetic-node geometry.

The two mechanisms are complementary. One is globally algebraic in the truncation index and exceptionally stable; the other is phase-sensitive and designed to separate a periodic asymptotic invariant. Both are sufficiently explicit to support immediate symbolic checking, high-precision computation, and a staged Lean formalization.

A Original audited TeX inventory

This appendix records the 57 distinct `.tex` paths included in the original repository-wide overlap audit. It is provenance for that audit, not a current corpus index: subsequent frontier reports increased the present tree to 67 TeX sources. Historical files are frozen versions and were treated as prior art even when their content has since been consolidated elsewhere.

A.1 Living top-level and canonical sources (6)

1. docs/FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex
2. docs/Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex
3. docs/Non_Elementarity_of_the_Fabius_Function/Non_Elementarity_of_the_Fabius_Function.tex
4. docs/fabius_lean_walkthrough/fabius_lean_walkthrough.tex
5. docs/non-formalized-research-frontiers/non-formalized-research-frontiers.tex
6. docs/non-formalized-research-frontiers/drafts/Thue_Morse_Formula_Atlas/Thue_Morse_Formula_Atlas.tex

A.2 Living Rvachev Fourier-decay drafts (12)

7. docs/non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay-2/Rvachev_Up_Fourier_Decay.tex
8. docs/non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay-3/Fourier_decay_of_Rvachev_up.tex
9. docs/non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Comparative_Audit/Rvachev_Up_Fourier_Decay_Comparative_Audit.tex
10. docs/non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Gentle_Guide/Rvachev_Up_Fourier_Decay_Gentle_Guide.tex
11. docs/non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Second_Wave_Audit/Rvachev_Up_Fourier_Decay_Second_Wave_Audit.tex
12. docs/non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/asymptotic-decay-rate-of-an-infinite-product-of-sinc-functions/asymptotic-decay-rate-of-an-infinite-product-of-sinc-functions.tex
13. docs/non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-1/rvachev_up_fourier_decay.tex
14. docs/non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-4/rvachev_up_fourier_decay.tex
15. docs/non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-5/rvachev_up_fourier_decay.tex
16. docs/non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-6/rvachev_up_fourier_decay.tex
17. docs/non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-7/rvachev_up_fourier_decay.tex

18. docs/non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-8/rvachev_up_fourier_decay.tex

A.3 Bundled source-paper TeX (7)

19. docs/papers/AIF_2017__67_2_637_0/AIF_2017__67_2_637_0-corrected.tex
20. docs/papers/AIF_2017__67_2_637_0/AIF_2017__67_2_637_0.tex
21. docs/papers/arXiv-1502.06738v2/Lacunary_products.tex
22. docs/papers/arXiv-1702.05442v1/09-Function.tex
23. docs/papers/arXiv-1702.06487v3/157-Arithmetic-v3.tex
24. docs/papers/arXiv-2211.13570v1/document.tex
25. docs/papers/ubiq15/ubiq15-corrected.tex

A.4 Frozen archive: early notes and standalone studies (5)

26. docs/archive/early-notes/FabiusAsymptotic/FabiusAsymptotic.tex
27. docs/archive/early-notes/K-foldsummationoverthesignedThue-Morsesequence/K-foldsummationoverthesignedThue-Morsesequence.tex
28. docs/archive/standalone-studies/Non_Elementarity_of_the_Fabius_Function/Non_Elementarity_of_the_Fabius_Function.tex
29. docs/archive/standalone-studies/Small_Argument_Asymptotics/Small_Argument_Asymptotics.tex
30. docs/archive/standalone-studies/fabius-inverse-dyadic-closed-form/fabius-inverse-dyadic-closed-form.tex

A.5 Frozen archive: primary expositions and supplements (8)

31. docs/archive/primary-exposition/Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex
32. docs/archive/primary-exposition/Fabius_Function_and_Rvachev_Up-2/Fabius_Function_and_Rvachev_Up.tex
33. docs/archive/primary-exposition/Fabius_Function_and_Rvachev_Up-3/Fabius_Function_and_Rvachev_Up.tex
34. docs/archive/primary-exposition/Fabius_and_Rvachev_Up-2/Fabius_and_Rvachev_Up.tex
35. docs/archive/primary-exposition/Fabius_and_Rvachev_up/Fabius_and_Rvachev_up.tex
36. docs/archive/primary-exposition/Fabius_function_and_up_function/Fabius_function_and_up_function.tex

- 37. docs/archive/primary-supplements/Fabius_Function_Missing_Parts/Fabius_Function_Missing_Parts.tex
- 38. docs/archive/primary-supplements/Fabius_Function_Missing_Parts-2/Fabius_Function_Missing_Parts.tex

A.6 Frozen archive: q -calculus, walkthrough, and glossary (6)

- 39. docs/archive/ q -calculus/Fabius_q_Special_Functions/Fabius_q_Special_Functions.tex
- 40. docs/archive/ q -calculus/fabius- q -special-functions/fabius- q -special-functions.tex
- 41. docs/archive/lean-walkthrough/Fabius_Function_Lean_Walkthrough/Fabius_Function_Lean_Walkthrough.tex
- 42. docs/archive/lean-walkthrough/fabius_lean_walkthrough/fabius_lean_walkthrough.tex
- 43. docs/archive/glossary/FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex
- 44. docs/archive/glossary/FabiusFunction_Mathematical_Glossary-2/FabiusFunction_Mathematical_Glossary.tex

A.7 Frozen archive: research frontiers (13)

- 45. docs/archive/research-frontiers/Fabius_Dyadic_Asymptotic_Bridge/Fabius_Dyadic_Asymptotic_Bridge.tex
- 46. docs/archive/research-frontiers/Fabius_Dyadic_Formulae_and_Alternative_Representations/Fabius_Dyadic_Formulae_and_Alternative_Representations.tex
- 47. docs/archive/research-frontiers/Fabius_Dyadic_Formulae_to_Asymptotics/Fabius_Dyadic_Formulae_to_Asymptotics.tex
- 48. docs/archive/research-frontiers/Fabius_Dyadic_q_Connections/Fabius_Dyadic_q_Connections.tex
- 49. docs/archive/research-frontiers/Fabius_Thue_Morse_Convergence_Rate/Fabius_Thue_Morse_Convergence_Rate.tex
- 50. docs/archive/research-frontiers/Fabius_Thue_Morse_Convergence_Rates/Fabius_Thue_Morse_Convergence_Rates.tex
- 51. docs/archive/research-frontiers/Primary_Exposition_Gap_Register/Primary_Exposition_Gap_Register.tex
- 52. docs/archive/research-frontiers/Repeated_Integration_and_Rvachev_Up/Repeated_Integration_and_Rvachev_Up.tex
- 53. docs/archive/research-frontiers/Rvachev_Up_from_Repeated_Integration/Rvachev_Up_from_Repeated_Integration.tex

- 54. docs/archive/research-frontiers/non-formalized-research-frontiers/non-formalized-research-frontiers.tex
- 55. docs/archive/research-frontiers/Thue_Morse_Formula_Atlas-1/Thue_Morse_Formula_Atlas.tex
- 56. docs/archive/research-frontiers/Thue_Morse_Formula_Atlas-2/Consolidated_Formula_Atlas_for_the_Thue_Morse_Sequence.tex
- 57. docs/archive/research-frontiers/Thue_Morse_Formula_Atlas-3/Thue_Morse_Formula_Atlas.tex

B Additional algebraic details

B.1 Derivation of the q -moment formula

Starting from [equation \(17\)](#),

$$\begin{aligned}
 S_{r,m}(q) &= \frac{1}{(q; q)_r} \sum_{j=0}^r (-1)^{r-j} q^{\binom{r-j+1}{2}} \begin{bmatrix} r \\ j \end{bmatrix}_q q^{jm} \\
 &= \frac{q^{rm}}{(q; q)_r} \sum_{k=0}^r (-1)^k q^{\binom{k+1}{2} - km} \begin{bmatrix} r \\ k \end{bmatrix}_q \\
 &= \frac{q^{rm}}{(q; q)_r} \sum_{k=0}^r q^{\binom{k}{2}} \begin{bmatrix} r \\ k \end{bmatrix}_q (-q^{1-m})^k \\
 &= \frac{q^{rm} (q^{1-m}; q)_r}{(q; q)_r}.
 \end{aligned}$$

For $m \geq r + 1$,

$$\begin{aligned}
 (q^{1-m}; q)_r &= \prod_{k=0}^{r-1} (1 - q^{1-m+k}) \\
 &= (-1)^r q^{\sum_{k=0}^{r-1} (1-m+k)} \prod_{k=0}^{r-1} (1 - q^{m-1-k}) \\
 &= (-1)^r q^{r(1-m) + r(r-1)/2} (q^{m-r}; q)_r.
 \end{aligned}$$

Multiplication by q^{rm} gives the exponent $r(r+1)/2$ in [equation \(20\)](#).

B.2 Why all transformed sinc-tail terms have one sign

The sign theorem is not a generic property of Richardson extrapolation. It relies on two positivity facts:

1. the log-sinc tail coefficients $d_m(t)$ are positive because $\zeta(2m) > 0$;
2. after cancellation, every geometric moment is

$$S_{r,m}(q) = (-1)^r \times \frac{q^{r(r+1)/2} (q^{m-r}; q)_r}{(q; q)_r},$$

whose remaining factors are positive for $m \geq r + 1$.

Their product therefore has the same sign for every residual degree. This is the mechanism behind the certified brackets.

B.3 Bell form of complete homogeneous functions

Taking logarithms of [equation \(72\)](#),

$$\log \prod_{j=0}^r (1 - a_j z)^{-1} = \sum_{k=1}^{\infty} \frac{z^k}{k} \sum_{j=0}^r a_j^k.$$

Comparing this with [equation \(6\)](#) gives

$$h_n(a_0, \dots, a_r) = \frac{1}{n!} \mathcal{B}_n(0!p_1, 1!p_2, \dots, (n-1)!p_n),$$

where $p_k = \sum_j a_j^k$. Substituting $a_j = (\lambda + j)^{-1}$ proves [equation \(76\)](#).

C Reproducibility notes

The numerical checks used arbitrary-precision arithmetic with the following direct definitions:

$$\begin{aligned} P_N(t) &= \prod_{k=1}^N \frac{\sin(t/2^k)}{t/2^k}, \\ P(t) &\approx \prod_{k=1}^{300} \frac{\sin(t/2^k)}{t/2^k}, \\ \log |\hat{P}_{N,r}(t)| &= \sum_{j=0}^r \lambda_{r,j}(1/4) \log |P_{N+j}(t)|. \end{aligned}$$

At 100-digit working precision, the 300-factor reference tail is far below the displayed errors. Symbolic checks verified the moment identities through $r = 8$ and multiple values of m before the general proofs were written.

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- [1] V. Reshetnikov, *The Fabius Function and Rvachev's Up-Function*, living primary exposition in the PROVEIT repository, Analysis/FabiusFunction/docs/Fabius_Function_and_Rvachev_Up/ (accessed 27 August 2026).
- [2] V. Reshetnikov, *Non-formalized Fabius Research Frontiers*, canonical quarantined frontier volume in the PROVEIT repository (accessed 27 August 2026).
- [3] V. Reshetnikov, *Thue–Morse Formula Atlas*, living draft in the PROVEIT repository (accessed 27 August 2026).
- [4] V. Reshetnikov et al., *Rvachev Up Fourier Decay dossier series*, twelve living TeX reports in the PROVEIT repository (accessed 27 August 2026).

- [5] J. Arias de Reyna, *An infinitely differentiable function with compact support: definition and properties*, Rev. Real Acad. Cienc. Exactas Fis. Nat. Madrid **76** (1982), 21–38; arXiv:1702.05442.
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- [7] R. M. Corless, G. H. Gonnet, D. E. G. Hare, D. J. Jeffrey, and D. E. Knuth, *On the Lambert W function*, Advances in Computational Mathematics **5** (1996), 329–359.
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- [11] J.-P. Allouche, M. Mendès France, and J. Peyrière, works on lacunary trigonometric products and the Thue–Morse sequence; source TeX included in the repository as `Lacunary_products.tex`.